Assignment – 12

Using the operators IN, ANY, and ALL

Write a query that selects all customers whose ratings are equal to or greater than ANY of Serres'.

```
mysql> select * from customers
    -> where rating >=
    -> (select min(rating) from customers
    -> where snum = (select snum from salespeople
    -> where sname = 'serres'));
+------+----------+----------+--------+------+
| Cnum | Cname    | City     | Rating | Snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.00 sec)
```

2) Write a query using ANY or ALL that will find all salespeople who have no customers located in their city.

```
mysql> select * from salespeople s
    -> where s.city <> All (
    -> select c.city from customers c
    -> where c.snum = s.snum);
+------+---------+-----------+------+
| Snum | Sname   | City      | Comm |
+------+---------+-----------+------+
| 1003 | Motika  | London    | 0.11 |
| 1007 | Rifkin  | Barcelona | 0.15 |
| 1004 | Axelrod | New York  | 0.10 |
+------+---------+-----------+------+
3 rows in set (0.04 sec)
```

3) Write a query that selects all orders for amounts greater than any for the customers in London.

```
mysql>  select * from orders
    ->  where amt > any
    -> (select amt from orders
    -> where cnum in
    ->  (select cnum from customers
    ->  where city = 'London'));
+------+---------+------------+------+------+
| Onum | Amt     | Odate      | Cnum | Snum |
+------+---------+------------+------+------+
| 3002 | 1900.10 | 1990-03-10 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-03-10 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-03-10 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-04-10 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-05-10 | 2006 | 1001 |
| 3010 | 1309.95 | 1990-06-10 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-06-10 | 2006 | 1001 |
+------+---------+------------+------+------+
7 rows in set (0.04 sec)
```

4) Write the above query using MIN or MAX

```
mysql> select * from orders
    -> where amt >
    -> (select min(amt) from orders
    -> where cnum in
    -> (select cnum from customers
    -> where city = 'London'));
+------+---------+------------+------+------+
| Onum | Amt     | Odate      | Cnum | Snum |
+------+---------+------------+------+------+
| 3002 | 1900.10 | 1990-03-10 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-03-10 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-03-10 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-04-10 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-05-10 | 2006 | 1001 |
| 3010 | 1309.95 | 1990-06-10 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-06-10 | 2006 | 1001 |
+------+---------+------------+------+------+
7 rows in set (0.00 sec)
```